

Physics Without Intuition

A Mechanical System for Solving Problems After the Flashcard Deck

Prologue

You have memorized 430 cards from [honors-physics-expression-to-equation-vocabulary.md](#). When a problem says "from rest," your hand writes $v_0 = 0$ before your eyes reach the next word (Card 1). When it says "collision," $\sum p_i = \sum p_f$ is already on the page (Card 72). When it says "spring," Hooke's Law (Card 42) and spring potential energy (Card 58) fire together.

You do not need this document to tell you which equation to use. The cards have already told you.

What you need is a **mechanical procedure for connecting cards**. A single physics problem fires 3, 5, or 10 cards across multiple domains. Your task is not to find equations — it is to trace every trigger phrase to its card, write the equation the card demands, chain the cards when domains cross, and verify every answer with dimensions. This document gives you that procedure.

Difficulty = number of cards fired. A Level-1 problem fires 3 cards. A Level-4 Olympiad problem fires 10. The method is identical either way.

Part 0: The Four Contracts

Before you touch a single equation, you make four commitments. These are not physics. They are contracts with yourself. They prevent sign errors, unit errors, and the most dangerous error of all — solving for the wrong variable.

§0.1 The Coordinate System

Fill in four blanks. Never change them mid-problem.

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+x direction: _____ (direction of primary motion)
+y direction: _____ (upward, or  $\perp$  away from incline surface)
Origin: _____ (object's starting position, or problem's
reference)
t = 0 at: _____ (instant action begins: release, throw,
collision)
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Rule 1 (+x): Point it where the object primarily goes. Car east $\rightarrow +x = \text{east}$. Block down incline $\rightarrow +x = \text{down the incline}$.

Rule 2 (+y): Point it upward. On an incline, point $+y$ perpendicular away from the surface — this makes the normal force purely $+y$ and decouples your equations.

- Rule 3 (origin):** Default: where the object starts. If the problem says "the ground," put the origin there.
- Rule 4 ($t = 0$):** The instant the action begins. Release. Throw. Collision.
- Golden rule:** Once written, never change. A negative answer means "opposite the + direction." It does not mean "wrong."

§0.2 Dimensional Analysis — Cards 394–430

Dimensional analysis is the only verification tool that works for every physics problem across every domain. It catches 80% of algebra mistakes. It requires zero understanding of the physics — only that you know the dimension of every quantity.

Base dimensions: $[M]$ (mass), $[L]$ (length), $[T]$ (time), $[Q]$ (charge), $[\Theta]$ (temperature).

Quantity	Definition	Dimension	Card
Speed v	$\Delta x / \Delta t$	$[L][T]^{-1}$	401
Acceleration a	$\Delta v / \Delta t$	$[L][T]^{-2}$	402
Force F	ma	$[M][L][T]^{-2}$	403
Energy E, K, U, W	$Fd, \frac{1}{2}mv^2$	$[M][L]^2[T]^{-2}$	404
Momentum p	mv	$[M][L][T]^{-1}$	405
Pressure P	F/A	$[M][L]^{-1}[T]^{-2}$	408
Frequency f	$1/T$	$[T]^{-1}$	416
Spring constant k	F/x	$[M][T]^{-2}$	415
Electric charge q	base	$[Q]$	419
Electric field E	F/q	$[M][L][Q]^{-1}[T]^{-2}$	420
Voltage V	U/q	$[M][L]^2[Q]^{-1}[T]^{-2}$	421
Magnetic field B	$F/(qv)$	$[M][Q]^{-1}[T]^{-1}$	424
Resistance R	V/I	$[M][L]^2[Q]^{-2}[T]^{-1}$	423

The dimension check (Card 430): After solving, compute the dimension of your answer. Does it match the dimension of the quantity you solved for? If not, your equation is wrong. Period. No exceptions. Fix it now, before you box the answer.

§0.3 The Variable Table

Before you think about any problem, draw this:

	Variable		Symbol		Value		Unit		Dimension		Source	
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Rules:

1. Every number from the problem gets a row. Include its unit.
2. The unknown gets ? in the Value cell.
3. **Implied zeros get rows.** "From rest" $\rightarrow v_0 = 0$ (Card 308). "Stops" $\rightarrow v = 0$ (Card 309). "Maximum height" $\rightarrow v_y = 0$ (Card 310). "Frictionless" $\rightarrow f = 0$ (Card 311). "Just loses contact" $\rightarrow N = 0$ (Card 316).
4. The Dimension column is your safety net. Your final answer must match it.

§0.4 The Trigger Trace

This is the core of the method. You do not "look up" equations. You recognize trigger phrases and fire cards. A **Trigger Trace** records every card that fired.

Extraction Algorithm

Execute these four passes mechanically:

1. **Read once.** No pencil. Gather the gist in five words or fewer.
2. **Extract numbers.** Every number with its unit $\rightarrow$ Variable Table (§0.3).
3. **Fire cards.** Underline every trigger phrase. For each one, write the card number and its instant equation. The set of cards that fire tells you the problem's domain.
4. **Identify the unknown.** Mark ? in the Variable Table.

Worked Extraction

"A stone is dropped from a 20 m high bridge. How long does it take to hit the water? ($g = 10 \text{ m/s}^2$)"

Trigger Phrase	Card	Instant Equation
"dropped"	1	$v_0 = 0$
$g = 10$, free-fall context	6	$a_y = -g = -10 \text{ m/s}^2$
"How long?"	7	t is the TARGET

Three cards fired. Domain: kinematics (Cards 1–22). No force words $\rightarrow$ no free-body diagram needed. The equation selector (Card 4) plus the missing-variable rule picks $\Delta y = v_{0y}t + \frac{1}{2}a_y t^2$ automatically.

Every worked example in this document begins with a Trigger Trace. It is the bridge between your flashcard training and problem-solving.

Part I: The I-SEE-R Protocol

Every problem in this document — single-domain or Olympiad synthesis — uses the same five-step protocol.

Step	Name	Action
I	Identify	Run the 4-pass extraction (§0.4). Produce: coordinate system, variable table, trigger trace.
S	Simplify	Cross out what does not matter. Frictionless? Massless rope? 1D motion? Ideal gas?
E	Equation	Write every equation from your trigger trace. Do not choose — the cards already chose. If multiple cards fired, write them all.
E	Execute	Substitute numbers. Solve algebraically. Keep symbols until the last possible moment.
R	Reflect	Three checks: (1) dimensions match (Card 430)? (2) one extreme case behaves correctly? (3) sign makes physical sense?

The **E — Equation** step used to be the hardest part of physics. For you, it is the easiest. The trigger phrase fires the card. The card demands the equation. You write it. The only remaining skill is chaining cards when a single problem spans multiple domains — and that is what Part V teaches.

Part II: Mechanics — Three Examples

Example 1: Free Fall

A stone is dropped from a 20 m high bridge. How long does it take to hit the water? ($g = 10 \text{ m/s}^2$)

Trigger Trace

Trigger Phrase	Card	Instant Equation
"dropped"	1	$v_0 = 0$
$g = 10$, free-fall context	6	$a_y = -g = -10 \text{ m/s}^2$
"How long?"	7	t is the TARGET

Cards fired: 1, 4, 6, 7. Domain: kinematics. No force words → no FBD.

Coordinate System

$+y$: upward | Origin: water surface ($y = 0$) | $t = 0$: release
→ $y_0 = +20 \text{ m}$, $y = 0$, $a_y = -10 \text{ m/s}^2$, $v_{0y} = 0$

Variable Table

Variable	Symbol	Value	Unit	Dimension
Initial position	y_0	20	m	$[L]$

Variable	Symbol	Value	Unit	Dimension
Final position	y	0	m	$[L]$
Initial velocity	v_{0y}	0	m/s	$[L][T]^{-1}$
Acceleration	a_y	-10	m/s ²	$[L][T]^{-2}$
Time	t	?	s	$[T]$

I-SEE-R

S — Simplify. 1D vertical motion. No air resistance. Acceleration constant at -10 .

E — Equation. Card 4 (kinematic suite) gives the master set. I know y_0, y, v_{0y}, a_y . I need t . I do not know v_y and the problem does not ask for it. The missing-variable rule (Card 4 table) says: need t , missing $v \rightarrow$ use $\Delta y = v_{0y}t + \frac{1}{2}a_yt^2$.

$$\Delta y = v_{0y}t + \frac{1}{2}a_yt^2$$

$$-20 = 0 \cdot t + \frac{1}{2}(-10)t^2$$

E — Execute. $-20 = -5t^2 \rightarrow t^2 = 4 \rightarrow t = 2.0$ s (positive root only — time cannot be negative).

R — Reflect.

- *Dimensions:* $[L] = [L][T]^{-2} \cdot [T]^2 = [L]$. Card 430: match. ✓
- *Extreme case:* Double the height $\rightarrow t = \sqrt{8} = 2.83$ s. Time scales as $\sqrt{h}$ — the signature of constant acceleration from rest. ✓
- *Independent check:* $v_y = (-10)(2) = -20$ m/s. $\bar{v}_y = (0 + (-20))/2 = -10$ m/s. $\Delta y = \bar{v}_yt = (-10)(2) = -20$ m. ✓

Answer: 2.0 s.

Example 2: Incline with Friction

A 5.0 kg block slides down a 30° incline. $\mu_k = 0.25$. Find the acceleration. ($g = 10$ m/s²)

Trigger Trace

Trigger Phrase	Card	Instant Equation
"slides down" (motion $\rightarrow$ kinetic)	29	$f_k = \mu_k N$
"incline" / 30°	38	$mg \sin \theta$ (along), $mg \cos \theta$ ($\perp$)
"friction" / μ_k	28, 29	$f_k = \mu_k N$, opposes motion
"Find acceleration"	23	$\sum F = ma$ is PRIMARY LAW

Cards fired: 23, 27, 28, 29, 38. Domain: forces. Multiple force words $\rightarrow$ FBD required.

Coordinate System

+x: down the incline | +y: $\perp$ away from surface
 $\rightarrow mg \sin\theta$ along +x | $-f_k$ along -x | +N along +y | $-mg \cos\theta$ along -y
 $\rightarrow a_y = 0$ (block stays on surface)

Variable Table

Variable	Symbol	Value	Unit	Dimension
Mass	m	5.0	kg	$[M]$
Angle	θ	30°	—	$[1]$
Kinetic friction coeff	μ_k	0.25	—	$[1]$
Gravity	g	10	m/s ²	$[L][T]^{-2}$
Acceleration	a	?	m/s ²	$[L][T]^{-2}$

I-SEE-R

S — Simplify. Block is sliding $\rightarrow$ kinetic friction, $f_k = \mu_k N$ (Card 29, not Card 30 which is static). Three forces: gravity, normal, friction. The block stays on the surface $\rightarrow a_y = 0$.

E — Equation. Card 23 ($\sum F = ma$) per axis:

$$+x: mg \sin \theta - f_k = ma$$

$$+y: N - mg \cos \theta = 0 \rightarrow N = mg \cos \theta \text{ (Card 27: normal force from } \sum F_y = 0)$$

Substitute Card 29 ($f_k = \mu_k N = \mu_k mg \cos \theta$) into the x -equation:

$$mg \sin \theta - \mu_k mg \cos \theta = ma$$

m cancels. Card 39 told us this happens for frictionless inclines; it happens with friction too:

$$a = g(\sin \theta - \mu_k \cos \theta)$$

E — Execute. $a = 10(0.5 - 0.25 \times 0.866) = 10(0.5 - 0.2165) = 2.84 \text{ m/s}^2$.

R — Reflect.

- $\mu_k = 0$: $a = 5.0 = g \sin \theta$. This is Card 39's result for a frictionless incline. ✓
- $\theta = 0$ (flat): $a = -2.5 \text{ m/s}^2$. Friction decelerates — correct, a block does not spontaneously slide on a flat surface. ✓
- $\theta = 90^\circ$ (vertical): $a = 10 = g$. No normal force, no friction, pure free fall. ✓

Answer: 2.84 m/s^2 down the incline.

Example 3: Perfectly Inelastic Collision

A 1500 kg car at 20 m/s east collides with a 2500 kg SUV at rest. They stick together. Find their velocity after.

Trigger Trace

Trigger Phrase	Card	Instant Equation
"collides"	72	$\sum p_i = \sum p_f$
"at rest"	1	$v_2 = 0$
"stick together"	73	$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f$

Cards fired: 1, 72, 73. Domain: momentum. Single collision, one equation, one unknown. KE is NOT conserved (Card 73 note) — and we do not need it.

Variable Table

Variable	Symbol	Value	Unit	Dimension
Car mass	m_1	1500	kg	$[M]$
SUV mass	m_2	2500	kg	$[M]$
Car initial v	v_1	20	m/s	$[L][T]^{-1}$
SUV initial v	v_2	0	m/s	$[L][T]^{-1}$
Final velocity	v_f	?	m/s	$[L][T]^{-1}$

I-SEE-R

S — Simplify. Perfectly inelastic (Card 73). One unknown, one equation. No external forces during the collision → momentum conserved (Card 72).

E — Execute. Card 73 gives the equation directly. Solve for v_f :

$$v_f = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{1500(20) + 0}{4000} = 7.5 \text{ m/s}$$

R — Reflect.

- *Dimensional:* $[M][L][T]^{-1}/[M] = [L][T]^{-1}$. ✓
- $m_2 = 0$ (no SUV): $v_f = 20 \text{ m/s}$. The car continues unchanged. The formula handles the edge case. ✓
- *Direction:* Positive = east. The wreckage moves east, slower than the car's original speed — physically correct. ✓
- *Energy note (optional):* Initial $K = 300,000 \text{ J}$, final $K = 112,500 \text{ J}$. Over 60% converted to heat and deformation — typical for perfectly inelastic collisions. ✓

Answer: 7.5 m/s east.

Part III: E&M — Two Examples

Example 4: Coulomb's Law

$q_1 = +3.0 \mu\text{C}$, $q_2 = -2.0 \mu\text{C}$, 0.10 m apart. Find the electrostatic force. ($k = 9.0 \times 10^9 \text{ N m}^2/\text{C}^2$)

Trigger Trace

Trigger Phrase	Card	Instant Equation
"charges" / " μC "	154	$F = k$
Opposite signs	156	Attractive force

Cards fired: 154, 156. Domain: electrostatics. Direct substitution — the card IS the answer.

I-SEE-R

S — Simplify. Point charges, vacuum, 1D radial. The charges have opposite signs → force is attractive (Card 156).

E — Execute. Card 154 fires:

$$F = 9.0 \times 10^9 \cdot \frac{(3.0 \times 10^{-6})(2.0 \times 10^{-6})}{(0.10)^2} = 5.4 \text{ N}$$

R — Reflect.

- *Dimensions:* $[k][Q]^2/[L]^2 = [M][L]^3[Q]^{-2}[T]^{-2} \cdot [Q]^2 \cdot [L]^{-2} = [M][L][T]^{-2} = \text{force}$ (Card 403).
✓
- *Doubled distance:* $F/4 = 1.35 \text{ N}$. Inverse square law — the signature of Coulomb (Card 154) and gravity (Card 94). ✓

Answer: 5.4 N, attractive.

Example 5: Compound Circuit

$R_1 = 4.0 \Omega$, $R_2 = 12.0 \Omega$ in parallel, $R_3 = 6.0 \Omega$ in series. 24 V battery. Find I_2 and V_2 .

Trigger Trace

Trigger Phrase	Card	Instant Equation
" Ω " / "resistor"	171	$V = IR$
"parallel" (R_1, R_2)	173	$1/R_{12} = 1/R_1 + 1/R_2$
"series" (R_{12}, R_3)	172	$R_{\text{total}} = R_{12} + R_3$
"24 V battery"	180	$V_{\text{terminal}} = \mathcal{E}$ (ideal)

Cards fired: 171, 172, 173, 175, 180. Domain: circuits. The strategy is reduction — Card 175.

I-SEE-R

S — Simplify. Ideal wires, ideal battery, DC steady state. Pure resistive circuit.

E — Execute. Card 175: reduce inward-outward.

Phase 1 — parallel (Card 173): $\frac{1}{R_{12}} = \frac{1}{4.0} + \frac{1}{12.0} = \frac{4}{12} \rightarrow R_{12} = 3.0 \, \Omega$

Phase 2 — series (Card 172): $R_{\text{total}} = 3.0 + 6.0 = 9.0 \, \Omega$

Phase 3 — total current (Card 171): $I_{\text{total}} = 24/9.0 = 2.667 \, \text{A}$

Phase 4 — distribute backward: $V_3 = I_{\text{total}} \cdot 6.0 = 16 \, \text{V}$. $V_{12} = 24 - 16 = 8 \, \text{V}$. Parallel branches share voltage (Card 173 note): $V_1 = V_2 = 8 \, \text{V}$.

$$I_2 = V_2/R_2 = 8/12 = 0.667 \, \text{A}$$

R — Reflect.

- KCL (Card 176): $I_1 = 8/4 = 2 \, \text{A}$. $I_1 + I_2 = 2.667 = I_{\text{total}}$. ✓
- Power (Card 178): $P_{\text{total}} = 24(2.667) = 64 \, \text{W}$. $P_1 + P_2 + P_3 = 16 + 5.33 + 42.67 = 64 \, \text{W}$. ✓

Answer: $I_2 = 0.667 \, \text{A}$, $V_2 = 8 \, \text{V}$.

Part IV: Thermodynamics & Waves

Example 6: The Combined Gas Law

Sealed container: $P_1 = 2.0 \, \text{atm}$, $V_1 = 5.0 \, \text{L}$, $T_1 = 300 \, \text{K}$. Compressed to $V_2 = 2.0 \, \text{L}$, heated to $T_2 = 450 \, \text{K}$. Find P_2 .

Trigger Trace

Trigger Phrase	Card	Instant Equation
"gas" / "pressure" / "volume" / "temperature"	217	$PV = nRT$
"sealed"	220	$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \, (n \text{ constant})$

Cards fired: 217, 220. Domain: thermodynamics. Sealed $\rightarrow n$ constant $\rightarrow$ combined gas law. Direct.

I-SEE-R

S — Simplify. Ideal gas (Card 217). Sealed container $\rightarrow n$ constant $\rightarrow R$ cancels (Card 220).

E — Execute. Card 220 fires. Solve for P_2 :

$$P_2 = P_1 \cdot \frac{V_1}{V_2} \cdot \frac{T_2}{T_1} = 2.0 \cdot \frac{5.0}{2.0} \cdot \frac{450}{300} = 2.0 \cdot 2.5 \cdot 1.5 = 7.5 \, \text{atm}$$

The factors reveal the physics: compression multiplies pressure by $2.5\times$. Heating multiplies it by $1.5\times$. Combined: $3.75\times$.

R — Reflect.

- *Isothermal* ($T_2 = T_1$): $P_2 = 5.0 \text{ atm}$ — Boyle's Law (Card 221). ✓
- *Isochoric* ($V_2 = V_1$): $P_2 = 3.0 \text{ atm}$ — Gay-Lussac's Law (Card 223). ✓
- Both factors multiply independently. The combined gas law (Card 220) reduces correctly to each special case. ✓

Answer: 7.5 atm.

Example 7: Standing Waves on a String

Guitar string: $L = 0.65 \text{ m}$, $\mu = 4.0 \times 10^{-3} \text{ kg/m}$, $F_T = 100 \text{ N}$. Find f_1 , f_3 , λ_3 .

Trigger Trace

Trigger Phrase	Card	Instant Equation
"string" / "tension"	136	$v = \sqrt{F_T/\mu}$
"fixed at both ends"	139	$f_n = nv/(2L)$
"fundamental frequency"	142	$f_1 = v/(2L)$
"third harmonic"	143	$f_3 = 3f_1$, $\lambda_3 = 2L/3$

Cards fired: 132, 136, 139, 142, 143. Domain: waves. Two-step: find v , then frequencies.

I-SEE-R

S — Simplify. Ideal string, uniform density, constant tension. Both ends fixed → displacement nodes at $x = 0$ and $x = L$ (Card 139).

E — Execute.

Card 136: $v = \sqrt{100/(4.0 \times 10^{-3})} = \sqrt{25,000} = 158.1 \text{ m/s}$

Card 142: $f_1 = v/(2L) = 158.1/1.30 = 121.6 \text{ Hz}$

Card 143: $f_3 = 3f_1 = 364.8 \text{ Hz}$, $\lambda_3 = 2L/3 = 0.433 \text{ m}$

R — Reflect.

- *Dimensions:* $\sqrt{[M][L][T]^{-2}/[M][L]^{-1}} = [L][T]^{-1} = \text{speed}$ (Card 401). ✓
- *Physical check:* Tighter string → higher v → higher f (sharper pitch). Heavier string → lower v → lower f (deeper pitch). Every string instrument confirms this. ✓
- *Harmonic spacing:* $f_n = nf_1$ — integer multiples. The defining property of a string fixed at both ends (Card 139). ✓

Answer: $f_1 = 122 \text{ Hz}$, $f_3 = 365 \text{ Hz}$, $\lambda_3 = 0.433 \text{ m}$.

Part V: The Bridge Protocol — Cards 321–375

The seven examples above are single-domain. One to four cards fire, one domain's laws apply, the answer comes directly. But SAT Physics and Olympiad problems chain domains. A bullet embeds in a block (momentum), then the block compresses a spring (energy). A gas expands (thermo), driving a piston (mechanics), which compresses a spring (SHM).

The vocabulary deck dedicates 55 cards to these chains: **Cards 321–375 (Cross-Domain Bridges)**. They do not contain new physics. They describe *architectures* — the structure of synthesis problems. Your job: recognize which bridge card your problem matches, then execute the chain.

The Bridge Recognition Algorithm

When a problem describes two or more distinct EVENTS separated in time, run this decision tree (Section 17.11 of the vocabulary deck):

Step	First event involves...	Bridge domain is...	Output is...	See cards...
1	COLLISION, EXPLOSION, RECOIL	MOMENTUM	v_f	321–327
2	POTENTIAL DIFFERENCE, E-FIELD	ENERGY ($\frac{1}{2}mv^2 = qV$)	v	338–344
3	FALLING, SLIDING, SPRING	ENERGY CONSERVATION	v or x	323, 328–337
4	GAS changing P, V, T	THERMO	P, V, T , or W	350–354
5	ACCELERATION over distance/time	KINEMATICS	v or t	—

After you identify the first domain and its output: that output is the *bridge variable* — it becomes an input to the second domain. The bridge card tells you the second domain's equation.

Worked Bridge Example: Bullet + Block + Spring

A 0.030 kg bullet at 500 m/s embeds in a 1.5 kg block on a frictionless surface. The block compresses a spring ($k = 2000$ N/m). Find max compression.

Trigger Trace

Trigger Phrase	Card	Instant Equation
"bullet embeds in block"	73	$m_b v_b = (M + m)v_f$
"frictionless surface"	31	$f = 0$
"compresses a spring"	42, 58	$F_s = -kx, U_s = \frac{1}{2}kx^2$
Bridge: bullet+block+spring	321	Momentum $\rightarrow v_f \rightarrow$ Energy

Cards fired: 31, 42, 58, 72, 73, 321. Two events $\rightarrow$ bridge architecture.

I-SEE-R

S — Simplify. Event 1: perfectly inelastic collision (Card 73). Event 2: spring compression on frictionless surface (Cards 31, 58, 60). The collision is fast — momentum is conserved but KE is NOT. After the collision, mechanical energy IS conserved (frictionless).

E — Equation. Card 321 gives the architecture. Two chained equations:

Domain A — Momentum (Card 73):

$$v_f = \frac{m_b v_b}{M + m} = \frac{0.030(500)}{1.53} = 9.80 \text{ m/s}$$

Domain B — Energy (Card 58):

$$\frac{1}{2}(M + m)v_f^2 = \frac{1}{2}kx_{\max}^2$$

$$x_{\max} = v_f \sqrt{\frac{M + m}{k}} = 9.80 \sqrt{\frac{1.53}{2000}} = 0.271 \text{ m}$$

R — Reflect.

- *Dimensions:* $[L][T]^{-1} \cdot \sqrt{[M]/([M][T]^{-2})} = [L][T]^{-1} \cdot [T] = [L]$. ✓
- *Key insight:* The bullet's initial KE is $\frac{1}{2}(0.03)(500)^2 = 3750 \text{ J}$. After embedding, the combined KE is only $\frac{1}{2}(1.53)(9.80)^2 = 73.5 \text{ J}$. Over 97% of the bullet's KE becomes heat during embedding — only **73.5 J** compresses the spring. Card 321's note warns you about this: "The bullet's initial KE is mostly converted to heat." ✓

Answer: 0.271 m (27.1 cm).

Part VI: The Variable-Group Method

Some problems resist direct mapping. The numbers describe the *middle* of a trajectory, but the question asks about the *beginning* or *end*. The given values and the target live in different "contexts" — different segments of the motion with different Δt , different Δx , but a shared boundary value.

This is not a physics problem. It is a bookkeeping problem. The Variable-Group Method solves it mechanically. No spatial reasoning. No trajectory visualization. Just tables and counting.

The Six Phases

Phase	Action	Mechanical Question
0	Recognize that grouping is needed	Can I write one equation containing BOTH all givens AND the target? If no → group.
1	Partition into contexts	"Which numbers describe the same thing?" Use the "About What" test.
2	Build mini tables	Each context gets its own variable table. Same symbol = same physical quantity.

Phase	Action	Mechanical Question
3	Find the richest context	Count KNOWN cells. Solve the context with the most knowns first.
4	Solve the richest context	Use the missing-variable rule (Card 4 kinematic selector).
5	Transfer the bridge	Does the variable just found appear in another context's table? Copy it there as KNOWN.
6	Repeat until target is KNOWN	Cycle: richest → solve → bridge → recount → richest → ...

The cycle terminates when the TARGET — the quantity the problem asks for — becomes KNOWN. You do not need to solve every context. You only need enough to reach the target.

Worked Example: The Window Problem

A ball is thrown straight up. It passes a **2.00 m** high window whose bottom is **7.50 m** above the ground. The ball takes **1.30 s** to travel from bottom to top of the window — and it is still going up. What was the ball's initial velocity at ground level? ($g = 10.0 \text{ m/s}^2$)

Trigger Trace

Trigger Phrase	Card	Instant Equation
"thrown straight up"	16	$a_y = -g, v_{0y} > 0$
"passes a point" / "crosses"	19	Middle-of-trajectory context — grouping required
"still going up"	13	v_y not yet zero at window top
$g = 10.0$	6	$a_y = -10.0 \text{ m/s}^2$

Cards fired: 4, 6, 13, 16, 19. Card 19 explicitly signals the Variable-Group Method.

Phase 0 — Recognize

Try writing one equation that connects **2.00 m**, **1.30 s**, **7.50 m** to the initial velocity at ground level. You cannot — the window numbers belong to a middle segment, the initial velocity belongs to the beginning. Grouping is required.

Phase 1 — Partition

Context	"About what?"	Variables
Context W (window)	Motion from window bottom to window top	$\Delta y_W = 2.00, t_W = 1.30, a_y = -10.0, v_{\text{bottom}}, v_{\text{top}}$
Context G (ground)	Motion from ground to window bottom	$\Delta y_G = 7.50, a_y = -10.0, v_{0y} \text{ (TARGET)}, v_{\text{bottom}}, t_G$

v_{bottom} appears in both — it is the **bridge variable**.

Phase 2 — Mini Tables

Context W: Known: $a_y, t_W, \Delta y_W$. Unknown: $v_{\text{bottom}}, v_{\text{top}}$.

Context G: Known: $a_y, \Delta y_G$. Unknown: v_{0y} (TARGET), v_{bottom}, t_G .

Phase 3 — Richest Context

Context W: 3 knowns. Context G: 2 knowns. → Solve Context W first.

Phase 4 — Solve Context W

Missing-variable rule (Card 4): need any variable; I do not need v_{top} → use $\Delta y = v_0 t + \frac{1}{2} a t^2$.

$$2.00 = v_{\text{bottom}}(1.30) + \frac{1}{2}(-10.0)(1.30)^2$$

$$2.00 = 1.30 v_{\text{bottom}} - 8.45$$

$$v_{\text{bottom}} = 8.04 \text{ m/s}$$

Phase 5 — Transfer

$v_{\text{bottom}} = 8.04$ is now KNOWN in Context G. Context G: 3 knowns ($a_y, \Delta y_G, v_{\text{bottom}}$).

Phase 4 again — Solve Context G

Need v_{0y} . Do not need t_G → Card 4: missing t → $v^2 = v_0^2 + 2a\Delta x$.

$$(8.04)^2 = v_{0y}^2 + 2(-10.0)(7.50)$$

$$64.6 = v_{0y}^2 - 150$$

$$v_{0y} = 14.7 \text{ m/s}$$

Phase 6 — Target Reached

TARGET $v_{0y} = 14.7 \text{ m/s}$ is KNOWN. Stop.

R — Reflect.

- *Plausibility:* $v_{0y} = 14.7$ at ground decreases to 8.04 at 7.50 m height. Gravity has decelerated the ball — consistent. ✓
- *Max height (Card 62):* $h_{\text{max}} = v_{0y}^2 / (2g) = 10.8 \text{ m}$. Window top is at 9.50 m, so the ball is indeed still rising at the window top — consistent with "still going up." ✓

Answer: 14.7 m/s upward.

Part VII: The Problem-Difficulty Ladder

Problems are not hard because you lack intuition. They are hard because they fire more cards. Difficulty is a countable property.

Level	Cards	Structure	Strategy
1. Direct	1–3	Single domain, all variables given	Trigger Trace → I-SEE-R. The card IS the equation.
2. Two-Step	2–5	One bridge variable	Chain: Domain A's output = Domain B's input. Find the bridge card (321–375).
3. Multi-Context	3–8	Numbers and target in different contexts	Variable-Group Method. Partition, count knowns, bridge, repeat.
4. Synthesis	5–15	3+ domains	Fire every card triggered by the problem. Cross out cards whose equations contain variables that are neither given nor reachable from other cards. What remains IS the solution path.

Your practice problem files are organized by level:

- [physics-questions.md](#) B1–B50: Level 1–2, 1–3 cards
- [physics-questions.md](#) I1–I50: Level 2–3, 2–5 cards
- [physics-questions.md](#) A1–A50: Level 3–4, 4–10 cards

Part VIII: Recovery Protocols — When You Are Stuck

Protocol A: Implicit Zero Scan (Cards 308–320)

Count knowns. Count unknowns. If unknowns exceed equations, re-read for implicit zeros. "From rest" → $v_0 = 0$ (Card 308). "Stops" → $v = 0$ (Card 309). "Maximum height" → $v_y = 0$ (Card 310). "Frictionless" → $f = 0$ (Card 311). "Just loses contact" → $N = 0$ (Card 316). "Massless rope" → T uniform (Card 312). "Constant speed" → $a = 0$ (Card 318).

Protocol B: Dimensional Back-Solve (Cards 394–430)

Write the dimension of the target. Write the dimensions of every given quantity. Is there only ONE way to combine the givens to produce the target's dimension? That combination IS the formula, up to a dimensionless constant. Card 430: "Do the dimensions of my answer match the dimension of what I am solving for?"

Protocol C: Extreme-Case Test

Set one variable to zero or infinity. Does the result make physical sense? $m_2 = 0$ in a collision → first object continues unchanged. $\mu_k = 0$ on an incline → $a = g \sin \theta$ (Card 39). $\theta = 0$ on an incline → flat surface behavior. A formula that fails any extreme case is wrong. Do not box it.

Protocol D: One-Sentence Summary

Stared for five minutes without writing? Stop. Describe the problem in plain language without physics words. "Something starts from rest, moves a distance while speeding up, I need its final speed." The physics terminology was the obstacle. Plain language reveals the structure.

Part IX: The Meta-Method — Learning New Physics

When you encounter a textbook chapter on a topic not yet in your flashcard deck:

- Pass 1 — Harvest equations.** Flip through. Copy every boxed, bold, or displayed equation into a list. Do not read the surrounding text.
- Pass 2 — Identify variables.** For each symbol in each harvested equation, write its name, its SI unit, and its dimension (derive from the equation using §0.2).
- Pass 3 — Classify.** Every physics equation answers one of four questions: Definition (what IS this quantity?), Force law (what force acts?), Motion law (how does it move?), or Conservation (what stays constant?).
- Pass 4 — Create flashcards.** Each new physics word → one new card in your personal deck, following the format of Cards 1–430. Now a future problem containing that word will mechanically trigger the new equation.

Quick-Reference Card

I-SEE-R Protocol

Step	Action
I	4-pass extraction: read → extract numbers → fire cards → identify unknown
S	Cross out what does not matter
E	Write every card equation from the trigger trace
E	Substitute numbers, solve algebraically
R	Dimension check (Card 430) + one extreme case + sign sanity

Trigger Trace Format

Trigger Phrase	Card	Instant Equation
"dropped"	1	$v_0 = 0$
"free fall"	6	$a = -g$
"How long?"	7	t is the TARGET

Bridge Decision Tree

Two or more events?

└ COLLISION/EXPLOSION/RECOIL first?

└ VOLTAGE/E-FIELD first?

└ FALLING/SLIDING/SPRING first?

└ GAS (P,V,T) first?

└ ACCELERATION first?

→ Momentum → v_f (Cards 321–327)

→ $\frac{1}{2}mv^2 = qV \rightarrow v$ (Cards 338–344)

→ Energy → v or x (Cards 328–337)

→ Thermo → P, V, T, W (Cards 350–354)

→ Kinematics → v or t

Equation Selection (Card 4)

Need v , missing t → $v^2 = v_0^2 + 2a\Delta x$

Need v , missing Δx → $v = v_0 + at$

Need Δx , missing v → $\Delta x = v_0t + \frac{1}{2}at^2$

Need Δx , missing a → $\Delta x = \frac{1}{2}(v_0+v)t$

Need t , missing v → $\Delta x = v_0t + \frac{1}{2}at^2$

Need t , missing Δx → $v = v_0 + at$

Implicit Zero Catalog (Cards 308–320)

Phrase	Zero	Card
"from rest", "dropped"	$v_0 = 0$	308
"stops", "comes to rest"	$v = 0$	309
"maximum height", "peak"	$v_y = 0$	310
"frictionless", "smooth"	$f = 0$	311
"just loses contact"	$N = 0$	316
"constant speed"	$a = 0$	318
"equilibrium"	$\sum F = 0, \sum \tau = 0$	319

The exam is won before you enter the room. Four hundred thirty cards have trained one reflex: see a phrase → write the equation. This document has trained a second: chain those equations mechanically, verify with dimensions, and never guess.

Difficulty equals the number of cards fired. A Level-1 problem fires three cards. A Level-4 Olympiad problem fires ten. The method is the same. Count the cards. Trace the triggers. Execute.

If you can calculate it, you have understood it.